



Winning Space Race with Data Science

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Outline

- Executive Summary
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- Methodology
- Results
- Conclusion
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Executive Summary

Method Summary

The goal of this work is to provide predictive capability for assessing when a SpaceX launch will or will not fail to land the 1st stage core

Publicly available data on SpaceX launches, covering over a decade, was acquired through REST API and web scraping. The data was cleaned and down selected to focus on the Falcon 9 series. Landing success was inspected through visualization and data query, to understand correlations in key launch metrics including launch site, payload and orbit. Classifier-type ML models were tested and optimized for building predictive capability.

Results Summary

SpaceX is continuing to improve its ability to land 1st stage cores. This is especially true as it progresses to larger payloads, mainly of the VLEO and ISS type orbits. These larger launches are correlated with localization of operations to Florida-based launch sites. A decision tree classification model approach can provide a 90%+ accurate model for predicting landing outcome.

Introduction

Project background and context

SpaceX is the leading, private space launch enterprise, with 100's of successful launches, including payload delivery to the International Space Station and deployment of multiple Starlink satellite arrays. As such, SpaceX is the key player in private space deployment, and any successful entrant must be able to deliver a superior deployment against SpaceX's legacy

Problems you want to find answers

SpaceX's profitability rests on its modular design, and ability to recover key assets, particularly the first stage launch core, allowing a single core to make multiple launches and bring down the total cost. Success of landing the first stage core is vital to recovery. As a strategy to compete against SpaceX, if it can be understood when a first stage core will fail to recover, this provides a niche in which an alternative, lower cost launch operator could bid for the contract.

There, we wish to understand SpaceX's first stage core recovery based on launch conditions to predict when recovery will succeed or fail.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
- Perform data wrangling
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Data Collection

Data was collected from the publicly available SpaceX API (version 4):
<https://api.spacexdata.com>

- Used python requests module

As well as obtained via the SpaceX Falcon 9 Heavy Launch Wikipedia page
https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches

- Used python beautifulsoup module with custom script to parse launch data table

Data Collection – SpaceX API

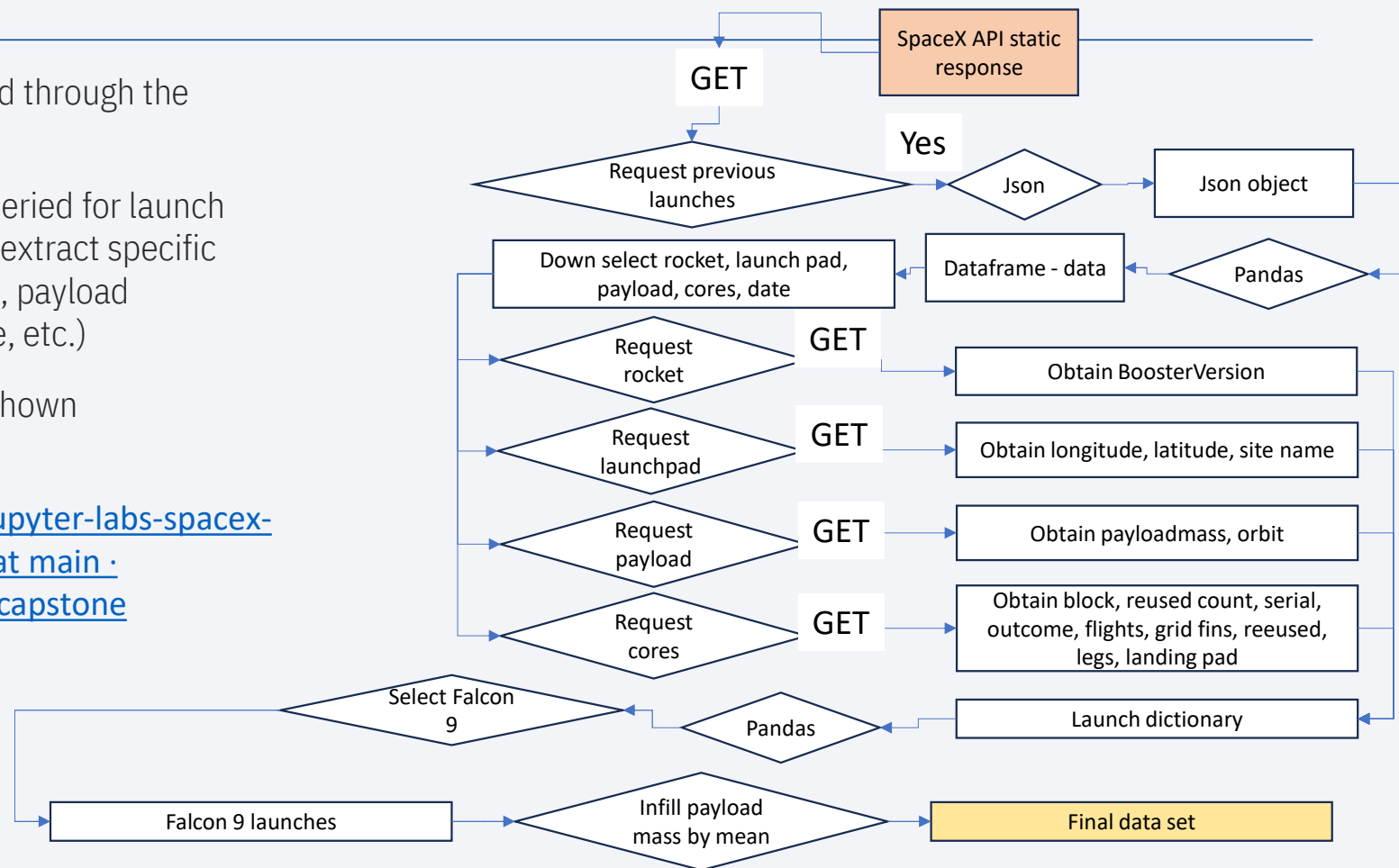
The SpaceX data was obtained through the publicly available API

Using python, this API was queried for launch data, which was then used to extract specific information (e.g. booster type, payload information, orbit, landing site, etc.)

The flow for this ingestion is shown

Github reference:

[se4_datascience_capstone/jupyter-labs-spacex-data-collection-api-v2.ipynb at main · bhwunsch/se4_datascience_capstone](#)



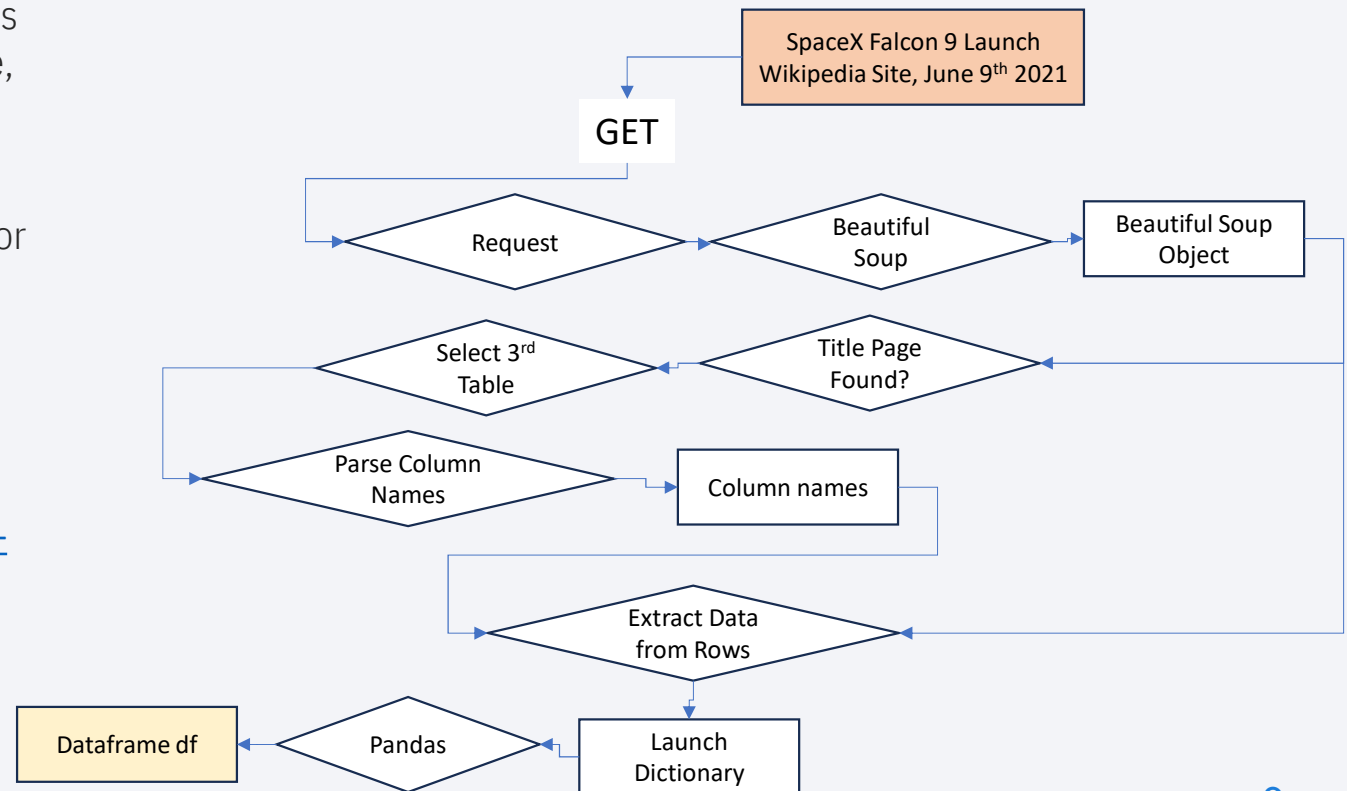
Data Collection - Scraping

Information on previous SpaceX launches is available publicly on the Wikipedia site, and this data helps in analyzing the success factors of first stage landing

Using python, the website was scraped for the key table, and this table was then systematically loaded and installed into

Github reference:

[se4_datascience_capstone/jupyter-labs-webscraping.ipynb](#) at main · [bhwunsch/se4_datascience_capstone](#)

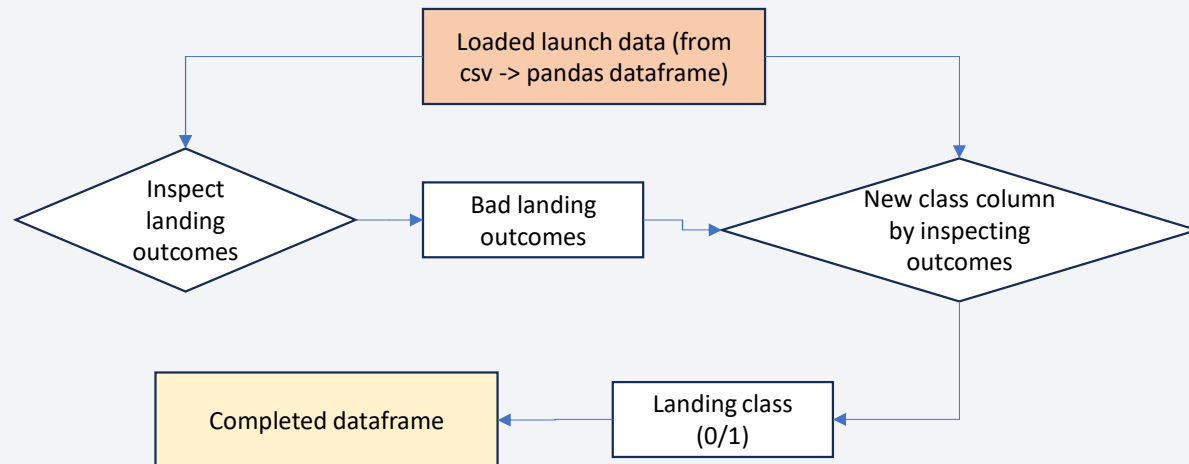


Data Wrangling

After data was collected using webscraping and the REST API from SpaceX, using python + pandas module, the data was analyze for any missing values, which were infilled with either the mean value (if numeric) or left empty in the case of landing site (which indicated none was used). The outcome of the landing was encoded as the *class* variable: 1 = success, 0 = failure, creating the finalized data set

Github reference:

[se4_datascience_capstone/labs-jupyter-spacex-Data wrangling-v2.ipynb](https://github.com/bhwunsch/se4_datascience_capstone/labs-jupyter-spacex-Data%20wrangling-v2.ipynb) at main · bhwunsch/se4_datascience_capstone



EDA with Data Visualization

The following visualizations were made to assess the trends affecting landing success

- Scatter - Payload mass vs flight number, shaded by landing success
- Scatter - Launch site vs flight number, shaded by landing success
- Scatter – Payload mass vs flight number, shaded by launch site
- Bar – Mean success (class) as function of orbit
- Scatter – Orbit type vs flight number, shaded by landing success
- Scatter – Orbit type vs payload mass, shaded by landing success
- Line – Average success of landing per year from 2010 - 2020

These visualizations helped to see the key features of launch site, payload and orbit type, and how they relate to the success rate of landing missions

Github reference:

[se4_datascience_capstone/jupyter-labs-eda-dataviz-v2.ipynb](https://github.com/se4_datascience_capstone/jupyter-labs-eda-dataviz-v2.ipynb) at main · bhwunsch/se4_datascience_capstone

EDA with SQL

The following queries were made to assess the SpaceX launch data:

- Get unique launch site
- Show first 5 launches that occurred at a site with “CCA” in its name
- Get total payload mass launched for NASA as a customer
- Get average payload mass carried by booster version F9 v1.1
- Get the date of the first successful ground landing
- Get the boosters which have successfully drone ship landed with payloads in the range of 4 to 6 Mg
- Get the total number of successful and failed landings
- List all booster versions which have carried the maximum launched payload
- List all records with failed drone ship landings in the year 2015, and show which booster version, landing site and month the launch occurred.
- List the total occurrences for each type of landing outcome between 2010/6/4 and 2017/3/20 in descending order

Github reference:

[se4_datascience_capstone/jupyter-labs-eda-sql-coursera_sqlite.ipynb](https://github.com/se4_datascience_capstone/jupyter-labs-eda-sql-coursera_sqlite.ipynb) at main · bhwunsch/se4_datascience_capstone

Build an Interactive Map with Folium

The location of each SpaceX launch site was mapped, using python + Folium, to gauge the geographic relationship between launch sites and successful stage 1 landings

The location of each launch site was marked with circles, combined with marker clusters to give the density of successful and failed landings at each site.

The distance from the launch site to the local coast was measured, indicating the proximity of the launch sites to the ocean, as is used for drone ship landings

Github reference: [se4_datascience_capstone/lab-jupyter-launch-site-location-v2.ipynb](https://github.com/se4_datascience_capstone/lab-jupyter-launch-site-location-v2.ipynb) at main · [bhwunsch/se4_datascience_capstone](https://github.com/bhwunsch/se4_datascience_capstone)

Build a Dashboard with Plotly Dash

Following interactions were created using Plotly Dash to display the launch data:

- Drop down menu to select launch site
- Pie chart for percentage successful landings for all launch sites – OR – percentage successful/failed landings for a selected launch site
- Slide bar to select launched payload weight (kg)
- Scatter plot depicting successful/failed landings (class = 1,0 respectively) for all launch sites or a selected one. Individual datum labelled by the type of booster used.

Key information to convey with these interactions:

- Ability to specify a launch site allows focusing on specific data
- Can readily observe the ratio of success/failure and how it is affected by the launch weight
- Can focus in on a specific payload range – crucial as this is the key parameter a client will supply
- Observe any correlations between booster type, launch site, and successful landing

Github reference: [se4_datascience_capstone/dashboard_script.py](https://github.com/se4_datascience_capstone/dashboard_script.py) at main · bhwunsch/se4_datascience_capstone

Predictive Analysis (Classification)

Scikit Learn ML models were used for the classification task of assessing successful/failed landing based on launch parameters

4 classifier models were attempted: support machine vector, decision tree, K-nearest neighbors, logistical regression

Each model was optimized using a grid search method, using a 10 fold split, to yield a performant parameter set. Each optimized model was tested and scored for accuracy

Confusion matrices were used to visually inspect the performance of the models and assess the best configuration.

Github reference: [se4_datascience_capstone/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb](https://github.com/se4_datascience_capstone/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb) at main · [bhwunsch/se4_datascience_capstone](https://github.com/bhwunsch/se4_datascience_capstone)

Results

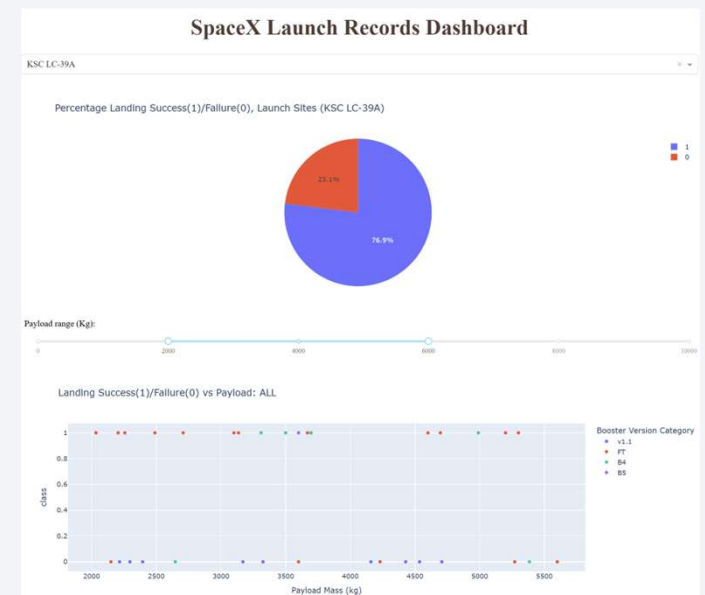
Exploratory data analysis results

- Launch site is important to landing success but not consistent or unilateral
- Payload is correlated to which launch site is used, and there is a rough trend towards larger payloads
- Orbit type affects landing success; while there are orbit types which have 100% success in landing, they are low sampled – the higher sampled orbit types have moderate success rates (~50-60%)
- SpaceX is improving its landing success over time; making it more difficult to expect a failed landing

Predictive analysis results

- Simple ML classification models can deliver in the ~80%+ accuracy rate
- Decision tree stood out as the only model capable of 87% accuracy, 90%+ scoring on test data, making it the best attempt to predict launch success
- For most models, the issue seems to be false positives – which may be due to the skew in successful/failed landing numbers

Interactive analytics demo in screenshots

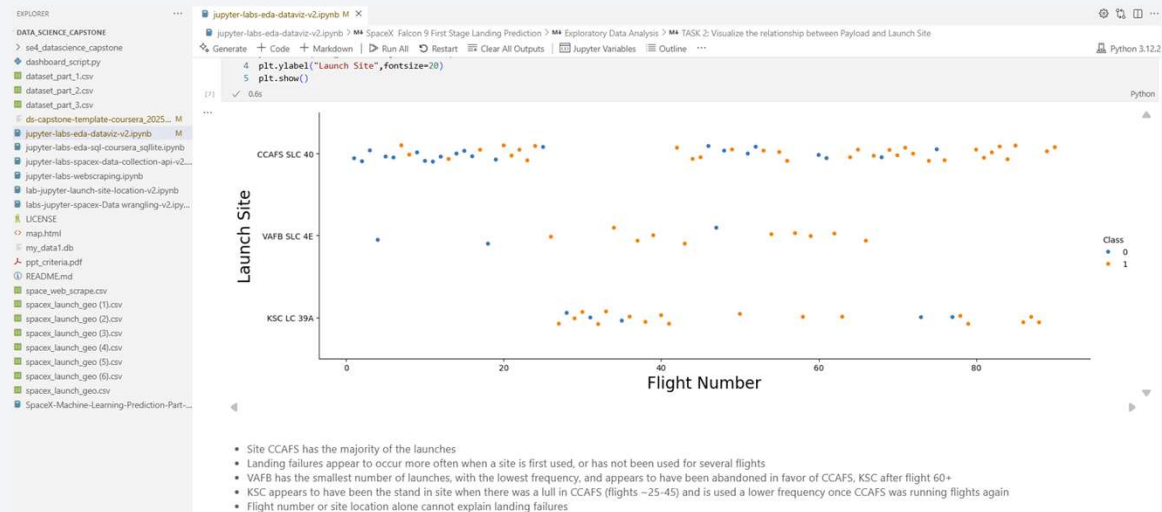
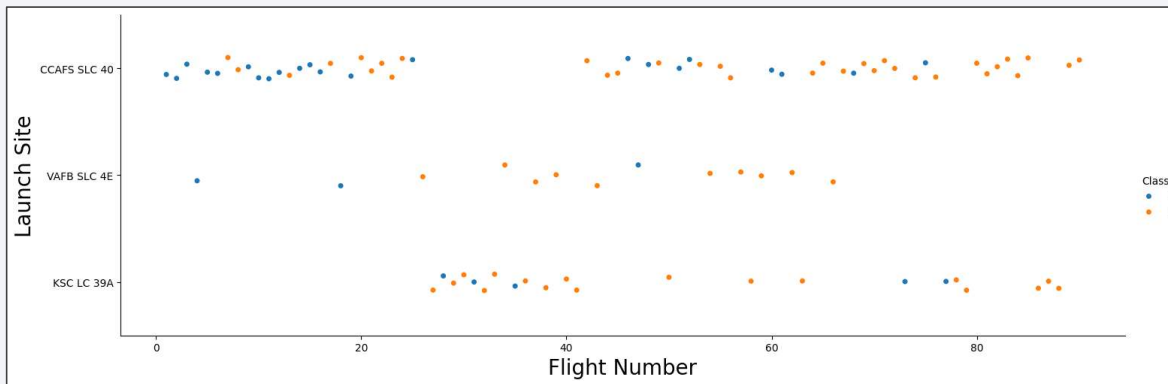


The background of the slide is a dynamic, abstract composition of numerous thin, overlapping lines and streaks in shades of blue, red, and cyan. These lines are oriented diagonally, creating a sense of movement and depth. The overall effect is reminiscent of a high-speed data visualization or a complex network diagram.

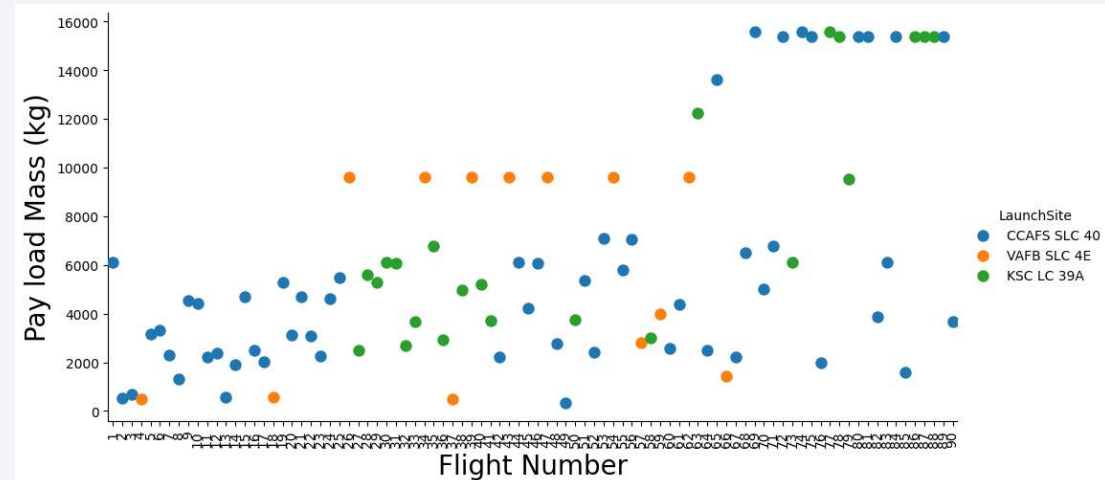
Section 2

Insights drawn from EDA

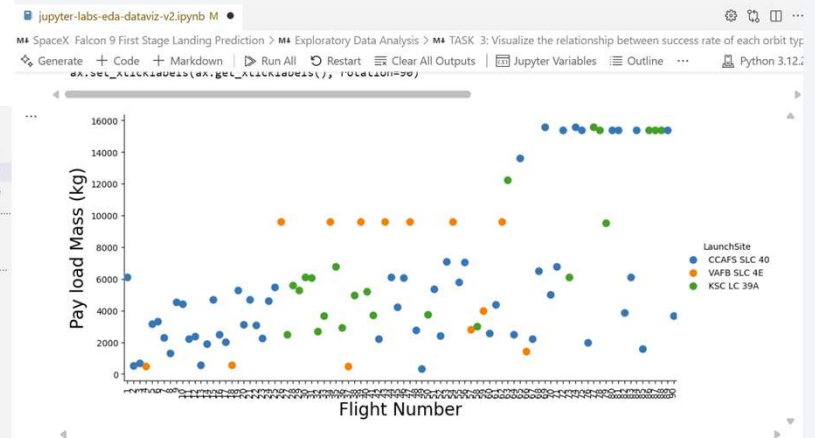
Flight Number vs. Launch Site



Payload vs. Launch Site

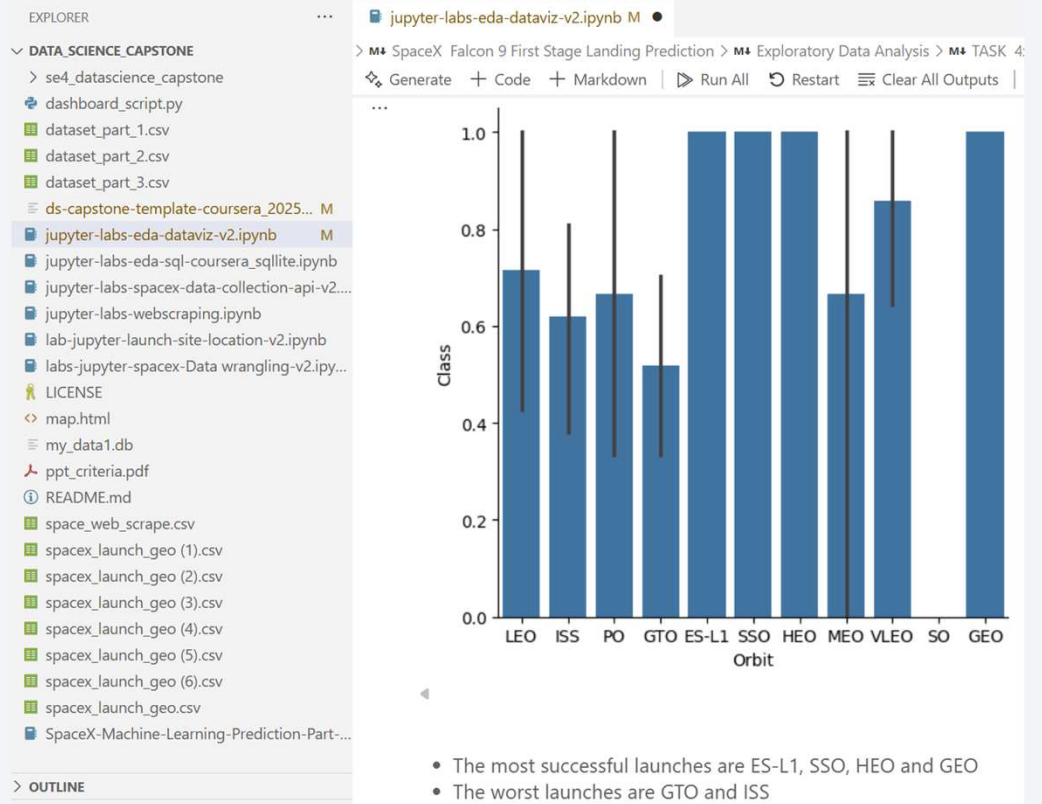
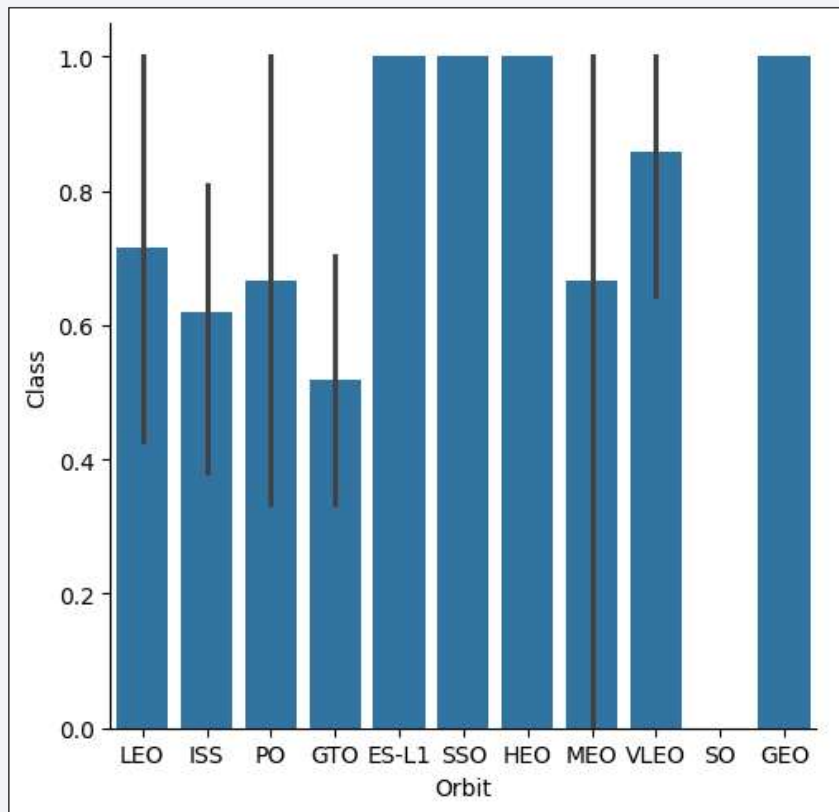


- dataset_part_2.csv
- dataset_part_3.csv
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- jupyter-labs-eda-dataviz-v2.ipynb M
- jupyter-labs-eda-sql-coursera_sqllite.ipynb
- jupyter-labs-spacex-data-collection-api-v2...
- jupyter-labs-webscraping.ipynb
- lab-jupyter-launch-site-location-v2.ipynb
- labs-jupyter-spacex-data-wrangling-v2.ipy...
- LICENSE
- map.html
- my_data1.db
- ppt_criteria.pdf
- README.md
- space_web_scrape.csv
- spacex_launch_geo (1).csv
- spacex_launch_geo (2).csv
- spacex_launch_geo (3).csv
- spacex_launch_geo (4).csv
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- spacex_launch_geo (6).csv
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- SpaceX-Machine-Learning-Prediction-Part...

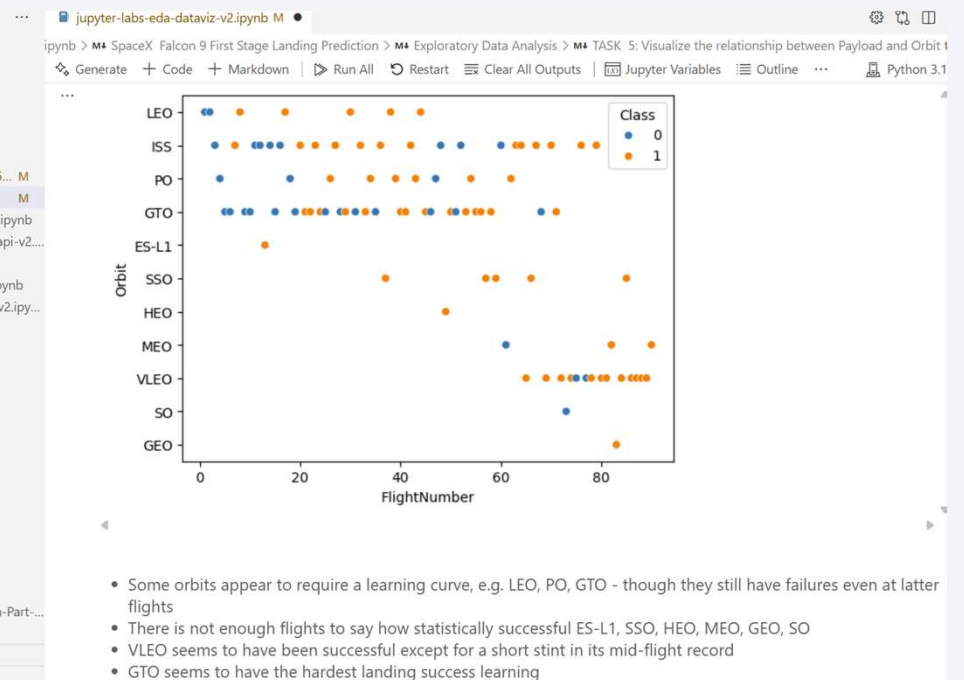
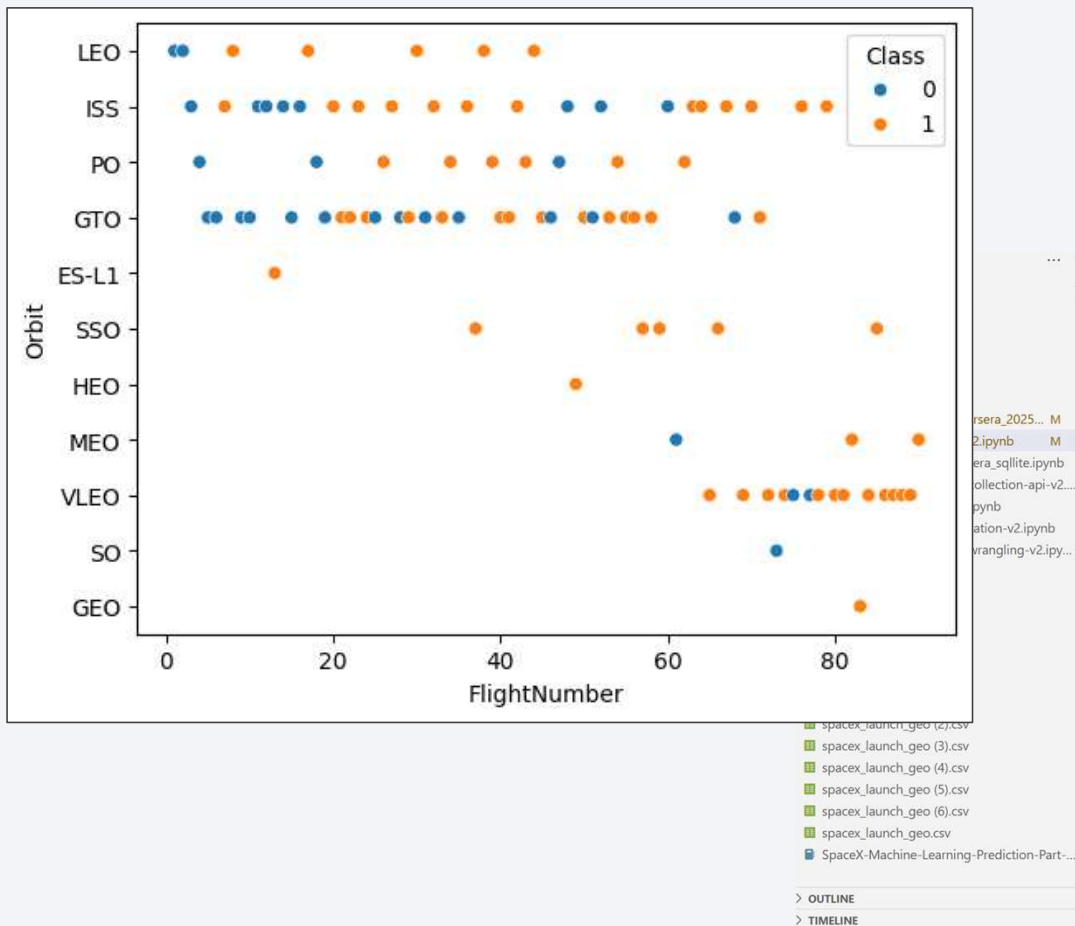


- There is a clear learning process of going from lower payload masses to higher over the flights
- CCFAS and KSC services mainly payloads < 6Mg and > 12Mg
- VAF8 seems to primarily service 2 main mass sizes: ~ 1Mg and ~10Mg

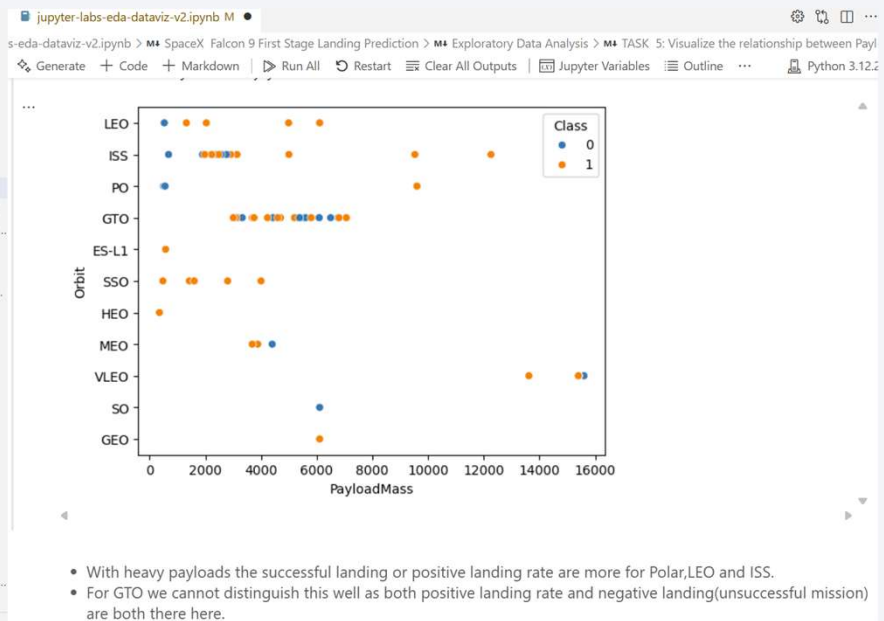
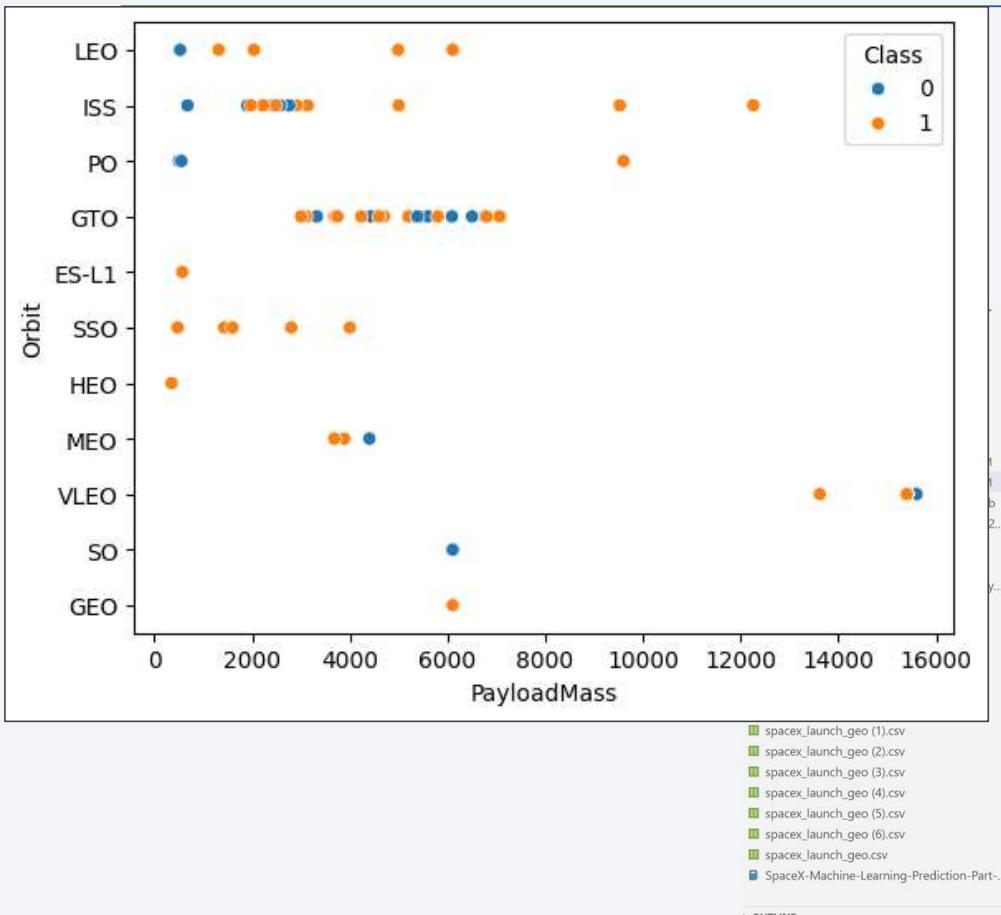
Success Rate vs. Orbit Type



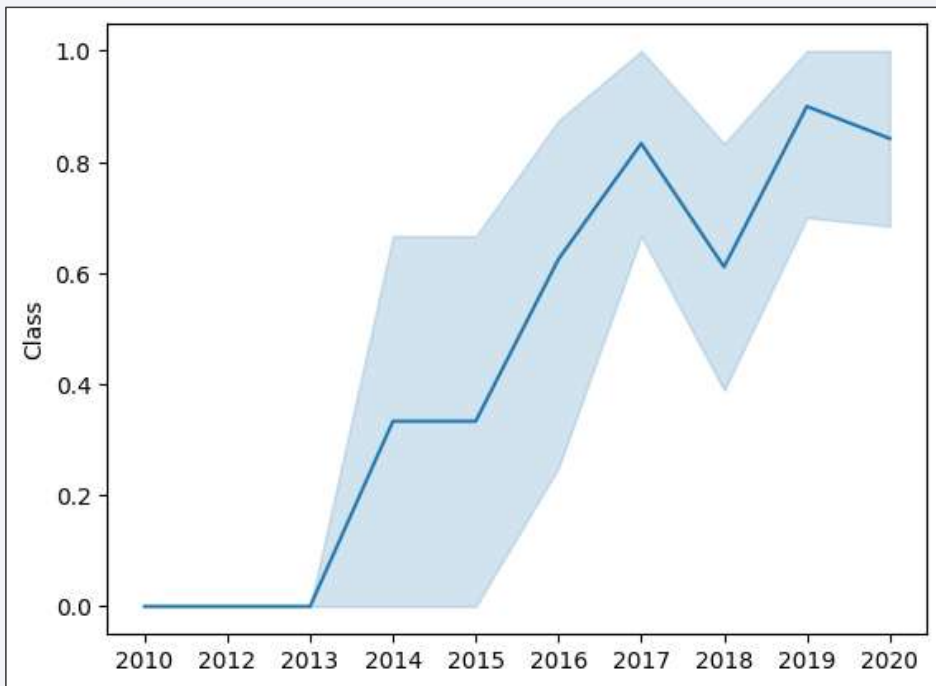
Flight Number vs. Orbit Type



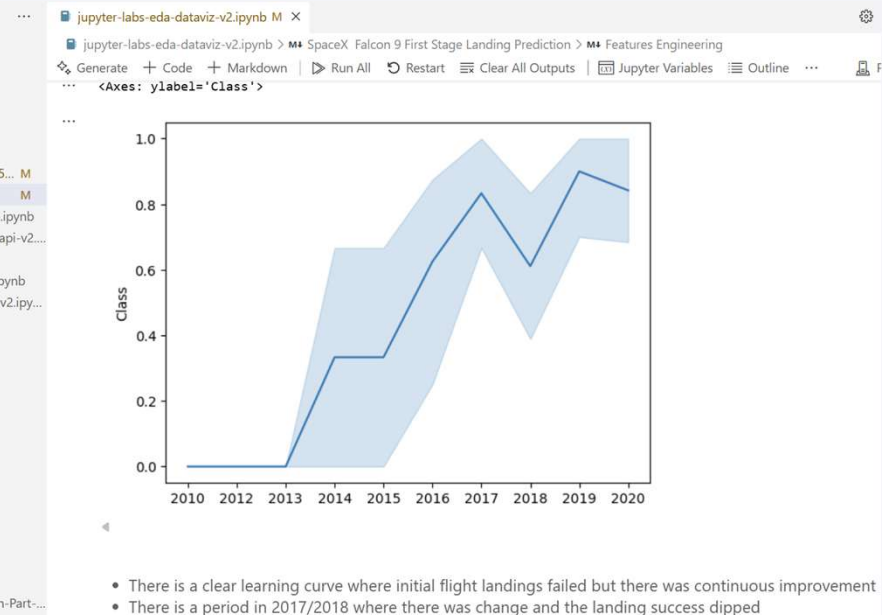
Payload vs. Orbit Type



Launch Success Yearly Trend



① README.md
■ space_web_scrape.csv
■ spacex_launch_geo (1).csv
■ spacex_launch_geo (2).csv
■ spacex_launch_geo (3).csv
■ spacex_launch_geo (4).csv
■ spacex_launch_geo (5).csv
■ spacex_launch_geo (6).csv
■ spacex_launch_geo.csv
■ SpaceX-Machine-Learning-Prediction-Part...



All Launch Site Names

SQL query of SpaceX launch data for unique launch site

```
1 %sql select distinct "Launch_Site" from SPACEXTABLE
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Launch Site Names Begin with 'CCA'

SQL query of SpaceX launch data for 5 launches that used a launch site starting with “CCA”, showing launches in 2010-2013 from the CCAFS LC-40 site

```
1 %sql select * from SPACEXTABLE where Launch_Site like '%CCA%' limit 5
✓ 0.0s
* sqlite:///my\_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677

Total Payload Mass

SQL query of SpaceX launch data for total payload mass (kg) of all launches carried out for NASA as a customer, showing over 45 Mg launched

```
1 %sql select sum("PAYLOAD_MASS_KG_") from SPACEXTABLE where Customer = 'NASA (CRS)'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

sum(PAYLOAD_MASS_KG_)
45596

Average Payload Mass by F9 v1.1

SQL query of SpaceX launch data for average payload mass (kg) of launches carried out with the F9 v1.1 booster

```
1 %sql select avg("PAYLOAD_MASS_KG") from SPACEXTABLE where Booster_Version = 'F9 v1.1'
✓ 0.0s

* sqlite:///my\_data1.db
Done.

avg(PAYLOAD_MASS_KG)
2928.4
```

First Successful Ground Landing Date

SQL query of SpaceX launch data giving earliest date, Dec 22nd 2015, for a successful ground landing

```
1 %sql select min("Date") from SPACEXTABLE where Landing_Outcome like '%Success (ground pad)%'
✓ 0.0s

* sqlite:///my\_data1.db
Done.
```

min(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL query of SpaceX launch data giving booster types which have successfully landed on drone ships, and were used with payloads in the range 4000-6000 kg

```
1 %%sql select "Booster_Version", "PAYLOAD_MASS_KG_"
2 from SPACEXTABLE
3 where Landing_Outcome like '%drone%'
4 and PAYLOAD_MASS_KG_ > 4000
5 and PAYLOAD_MASS_KG_ < 6000
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

Booster_Version	PAYLOAD_MASS_KG_
F9 FT B1020	5271
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Total Number of Successful and Failure Mission Outcomes

SQL query of SpaceX launch data showing number of successful/failed landings

```
1 %%sql
2 SELECT 'Success' AS outcome, COUNT(*) AS total
3 FROM SPACEXTABLE
4 WHERE Mission_Outcome LIKE '%Success%'
5 UNION ALL
6 SELECT 'Fail' AS outcome, COUNT(*) AS total
7 FROM SPACEXTABLE
8 WHERE Mission_Outcome LIKE '%Fail%';
9
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

outcome	total
Success	100
Fail	1

Boosters Carried Maximum Payload

SQL query of SpaceX launch data giving all booster types which have carried the maximum launched payload mass of 15.6 Mg

```
1 %%sql select "Booster_Version", PAYLOAD_MASS_KG_ from SPACEXTABLE
2 where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) as max_load from SPACEXTABLE)
✓ 0.0s
```

* [sqlite:///my_data1.db](#)
Done.

Booster_Version	PAYLOAD_MASS_KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

SQL query of SpaceX launch data giving failed drone ship landings in 2015 by their month, booster type and which site they were launched from

```
1 %%sql select substr(Date, 6,2) as month, "Landing_Outcome", "Booster_Version", "Launch_Site"
2 from SPACEXTABLE
3 where Landing_Outcome like '%Failure (drone ship)%'
4 and substr(Date,0,5)='2015'
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL query of SpaceX launch data between June 4th 2010 and March 20th 2017, showing the landing outcomes for all landing types used

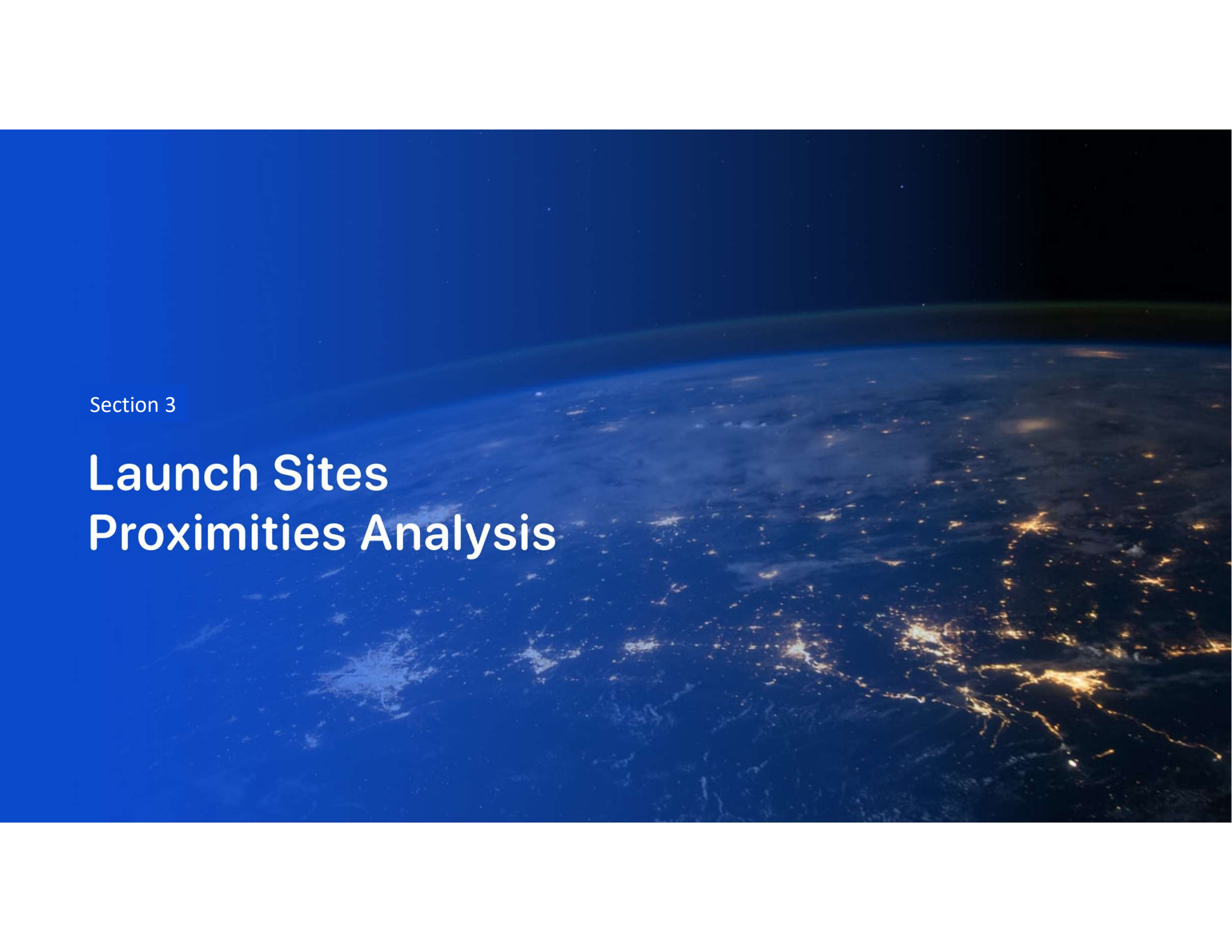
```
1 %%sql select Landing_Outcome, count(Landing_Outcome) as total_outcomes from SPACEXTABLE
2 where Date > "2010-06-04"
3 and Date < "2017-03-20"
4 group by Landing_Outcome
5 order by total_outcomes desc
```

✓ 0.0s

* [sqlite:///my_data1.db](#)

Done.

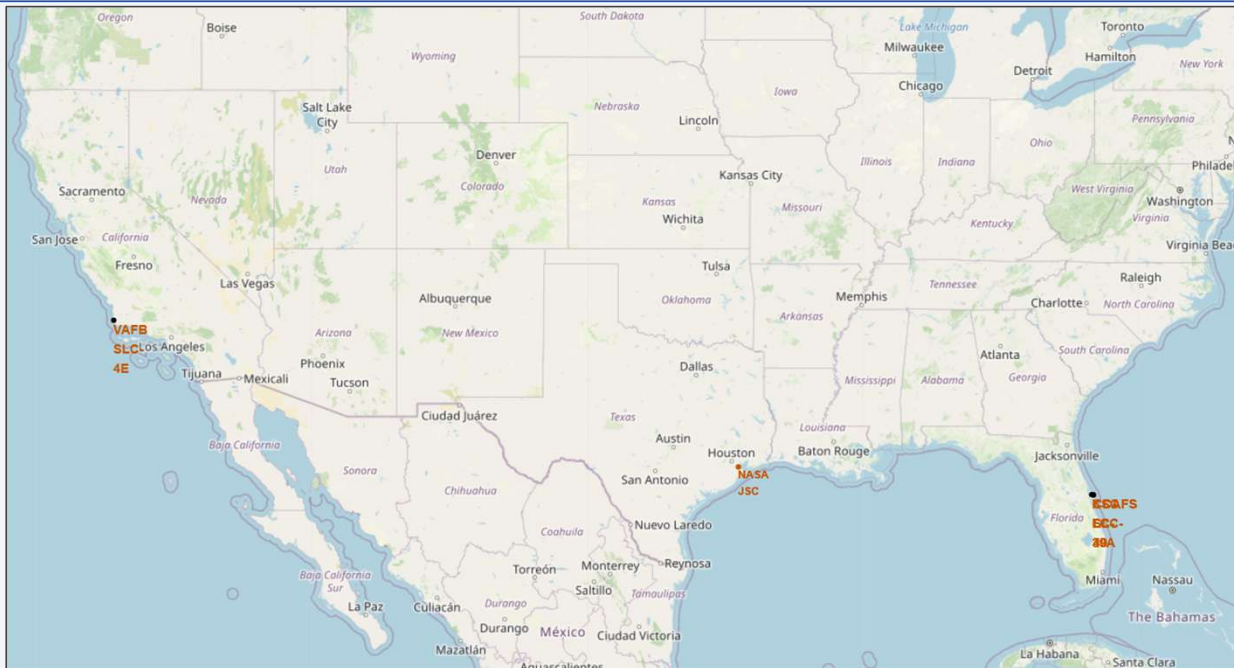
Landing_Outcome	total_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a deep blue, with a thin white line representing the horizon. Below the horizon, the Earth's surface is visible, with numerous bright yellow and orange lights indicating urban areas. The lights are concentrated in the lower right portion of the image, with some scattered lights visible elsewhere. The overall scene is a high-contrast, high-resolution view of the planet from space.

Section 3

Launch Sites Proximities Analysis

Geographic Locations of SpaceX Launch Sites

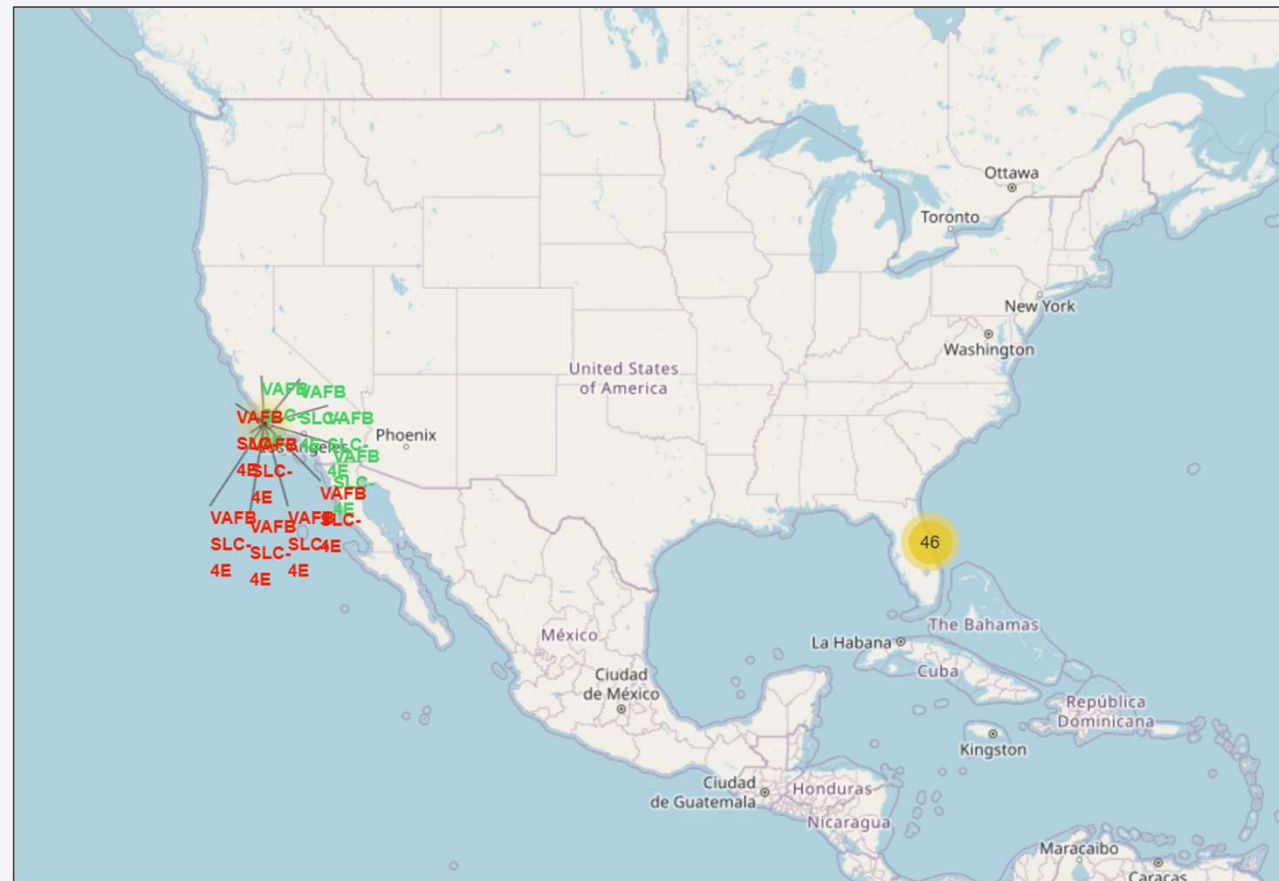


The figure shows the location of all launch sites used by SpaceX in this study of landing success. It is clear all sites are located in the southern United States, near coastal facilities. This positions the majority of the sites as close to the equator as possible from the continental United States

Geographic Distribution of Successful Landings

The figure shows an example of using Folium mapping to highlight landing status

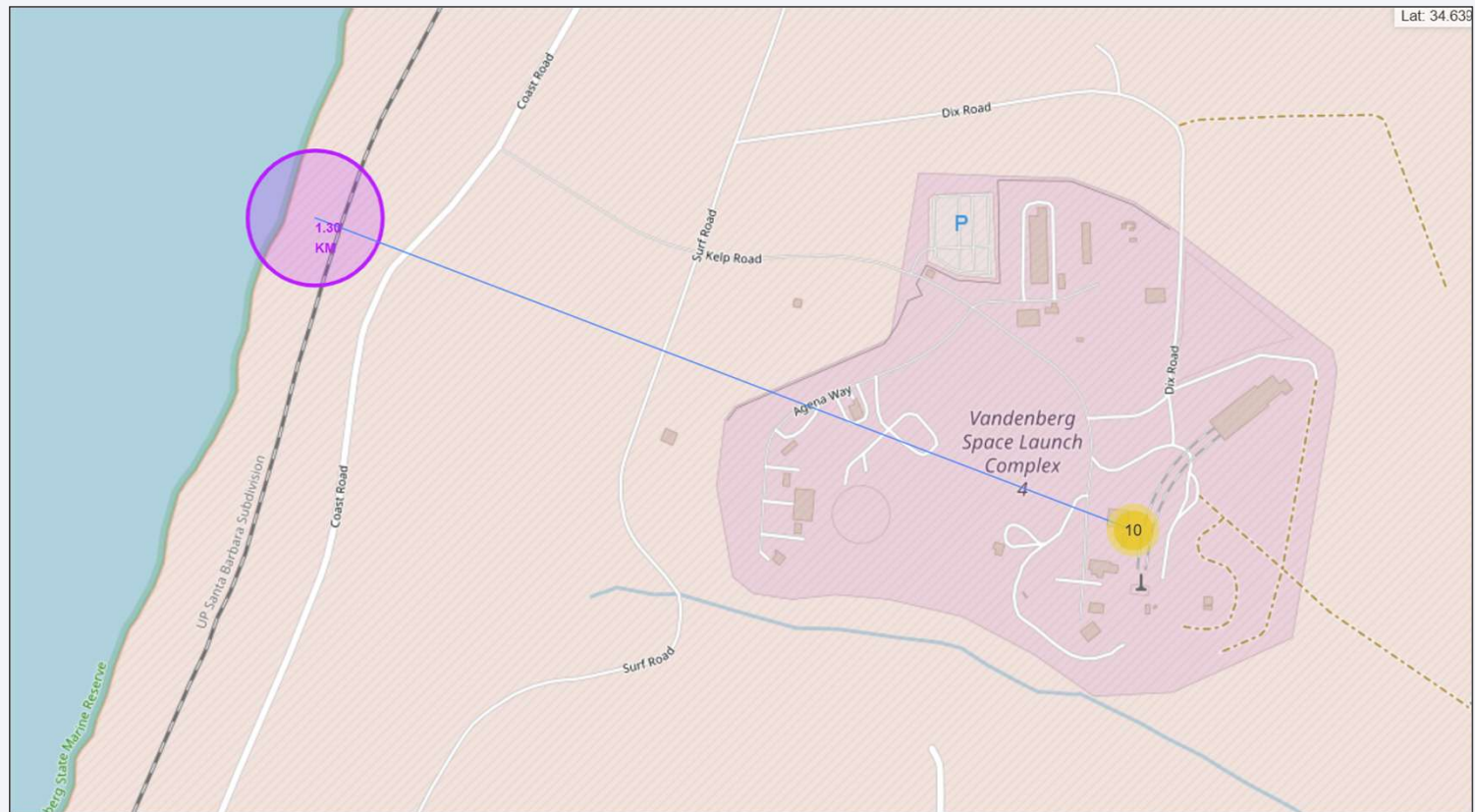
The successful (green) and failed (red) landings are highlighted for the Vandenberg Space Force Base, outside Los Angeles, California, showing that a majority of launches failed to land

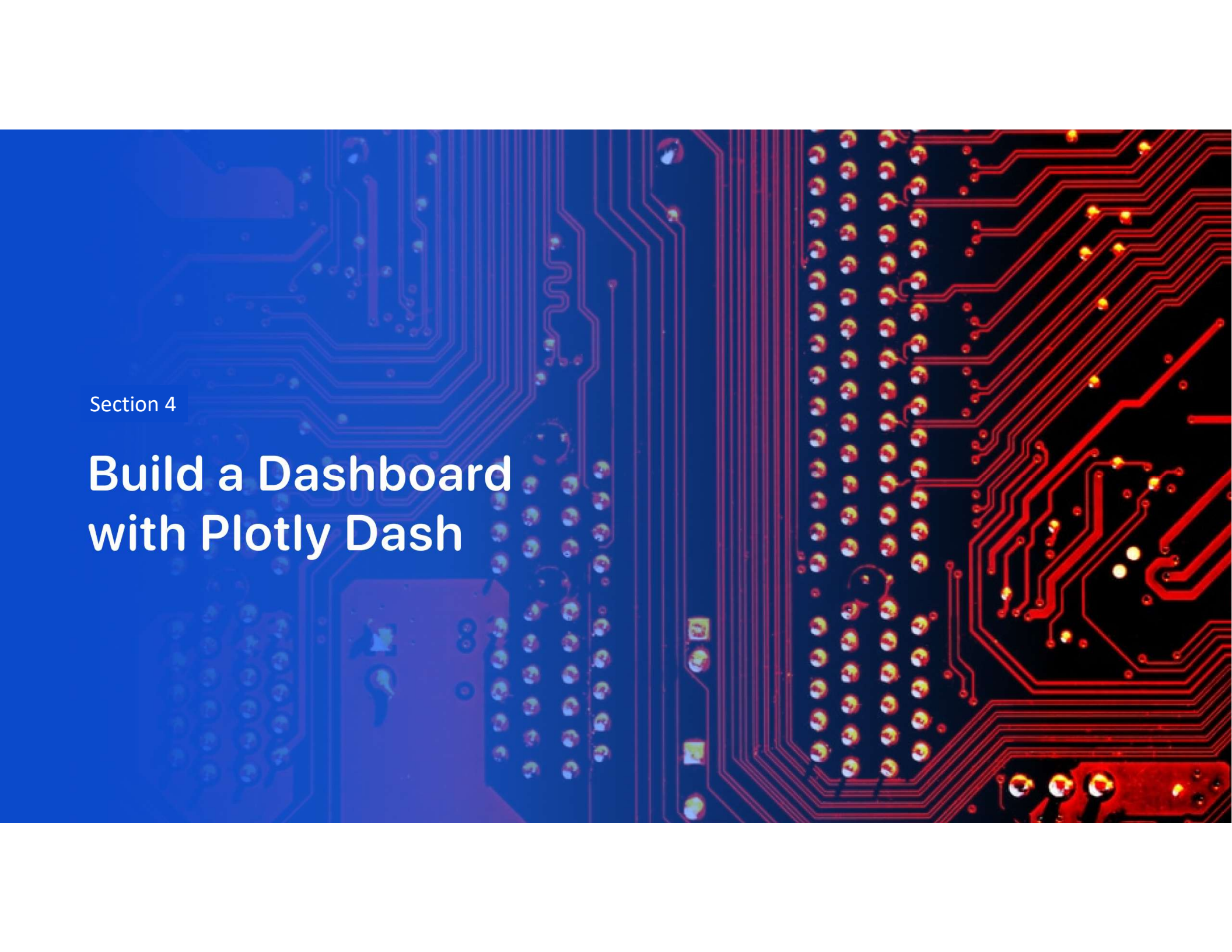


Geographic Detail of Launch Site Position

The figure shows an example of using Folium mapping to highlight geographic distances from SpaceX launch sites

The example shows the distance from the Vandenberg Space Force Launch site to the Pacific coast, indicating the proximity of the coastline and suggesting drone ships as the primary landing mode

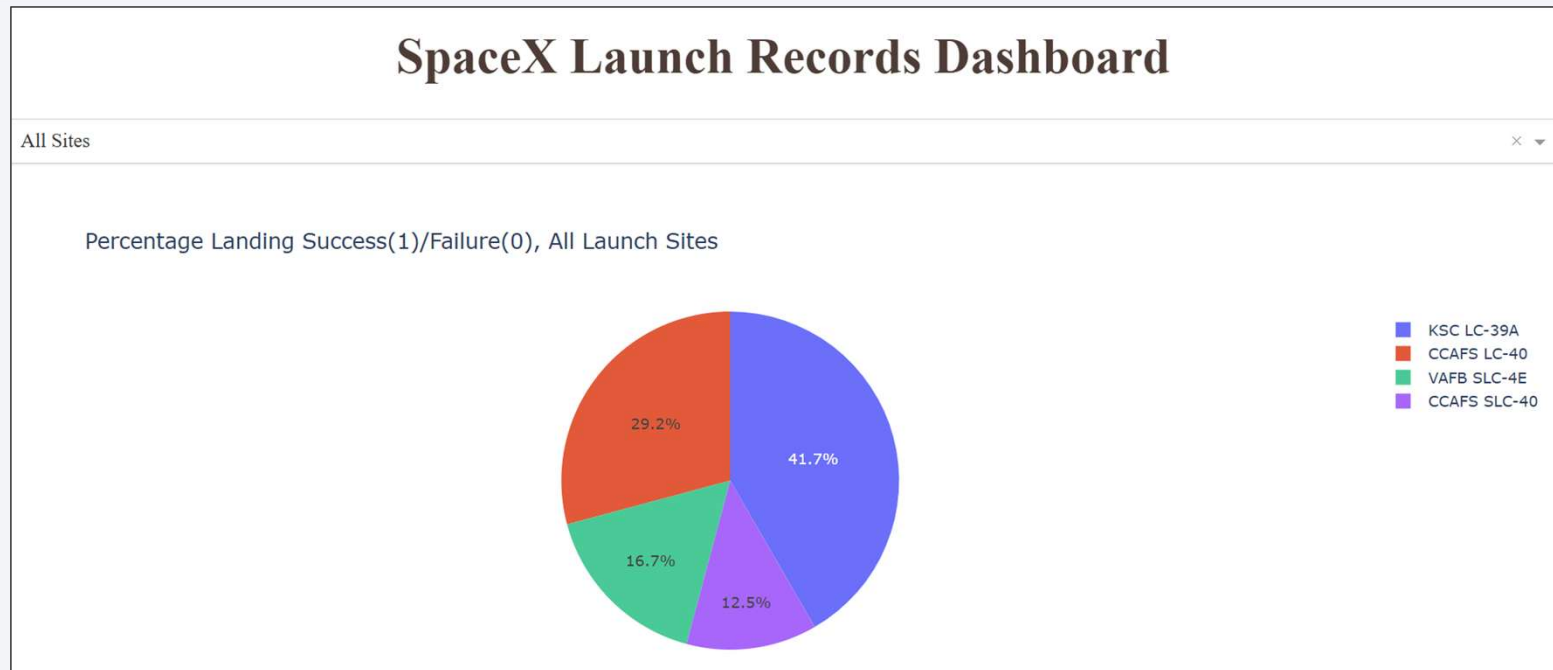




Section 4

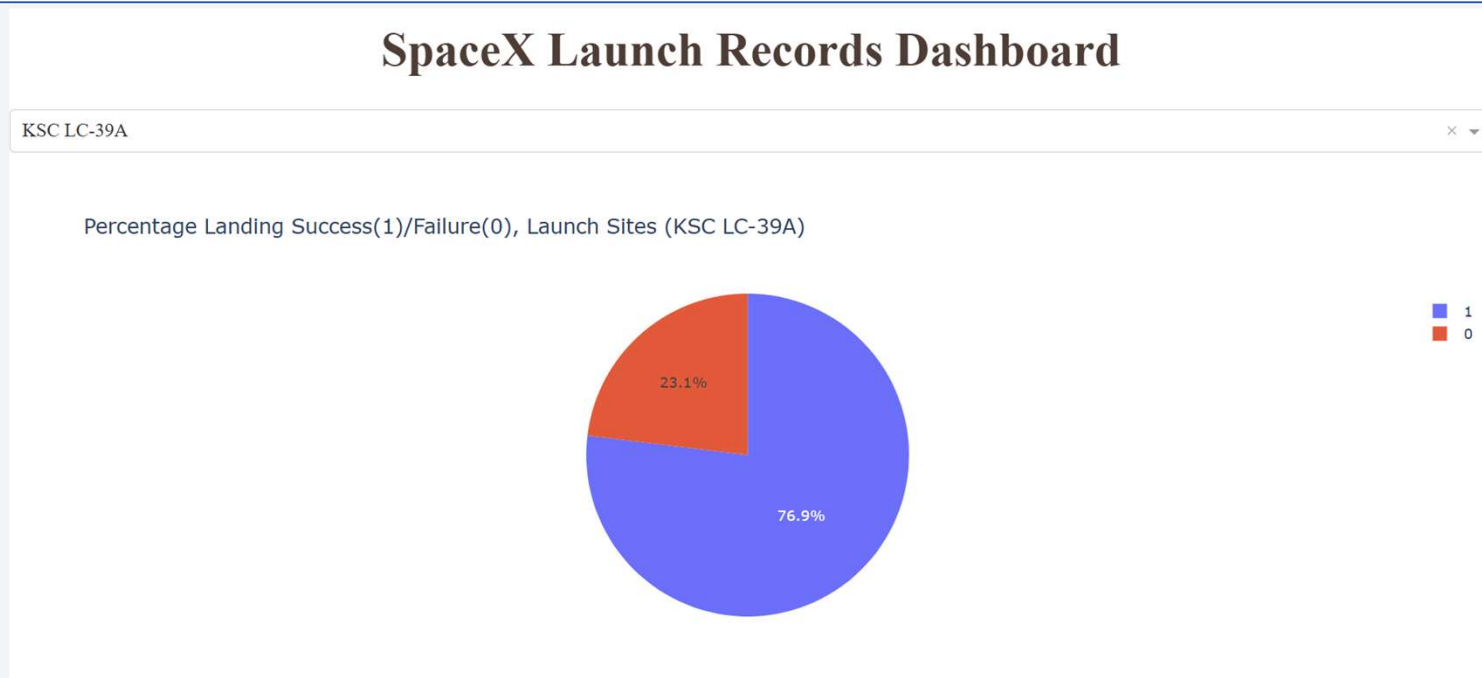
Build a Dashboard with Plotly Dash

SpaceX: Percentage Successful Landings for all Launch Sites



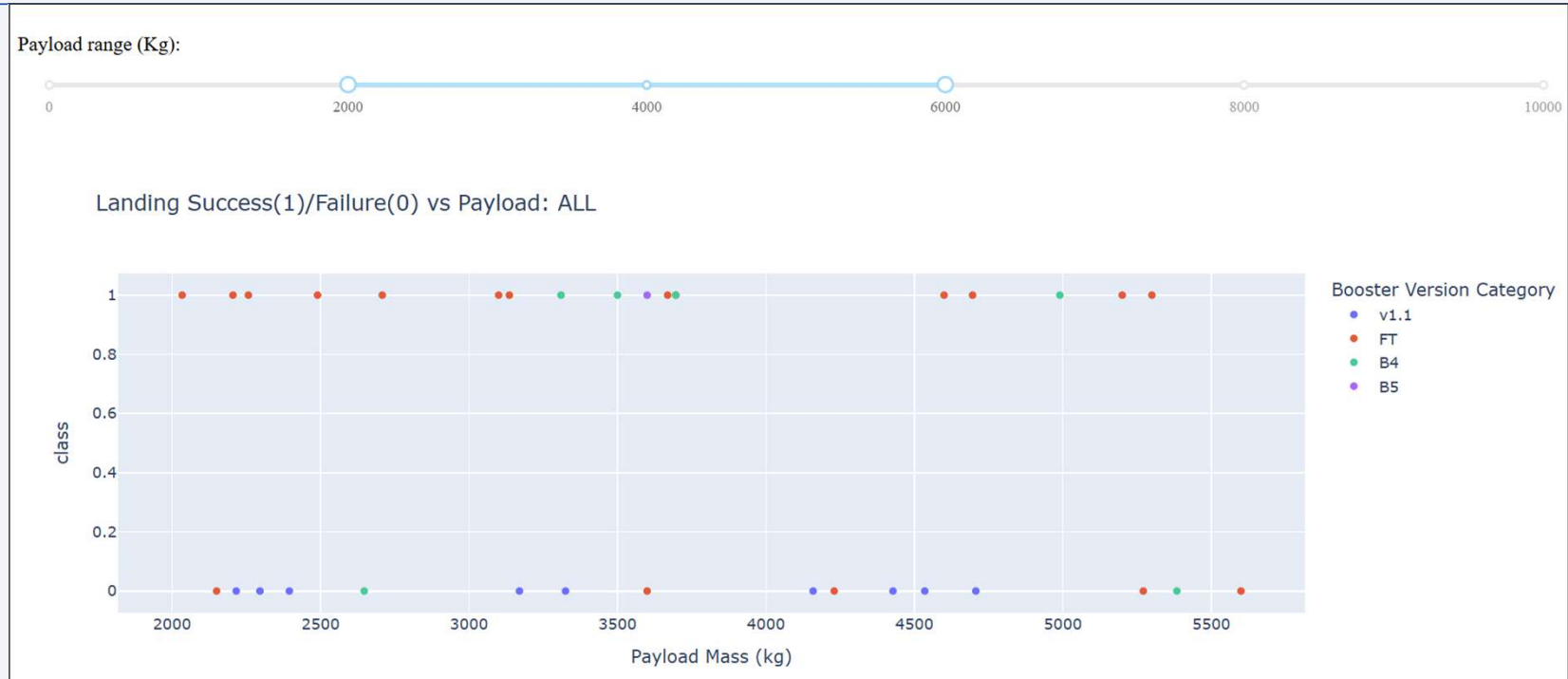
The figure shows the percentage of successful landings for each launch site used by SpaceX. The highest success is seen for the KSC LC-39A site, and lowest for the CCAFS SLC-40. It is noteworthy that all sites have <50% success rate for landing

SpaceX: Percentage Successful/Failed Landings for the most successful Launch Site: KSC LC-39A



The figure shows the percentage of successful/failed landings for site KSC LC-39A; the most successful site for landings. KSC achieves successful landings greater than 3 out of 4 launches

SpaceX: Landing Success/Failure as function of Launched Payload Weight



The figure shows the percentage of successful/failed landings (noted as *class*, with success = 1, and failure = 0) as a function of payload weight (kg) for all launch sites studied. The data have been shaded according to the booster type used for the launch. The payload range has been scaled between 2000 – 6000 kg to resolve the largest concentration of launched payloads. It is seen that in this range, there is very little observed correlation between successful launch and payload weight

The background of the slide features a dynamic, abstract image. On the left, there is a solid blue area. To the right, a perspective view of a tunnel is shown, with its walls and floor curving into the distance. The tunnel's interior is illuminated with a mix of blue and white light, creating a sense of depth and movement. The overall aesthetic is modern and technological.

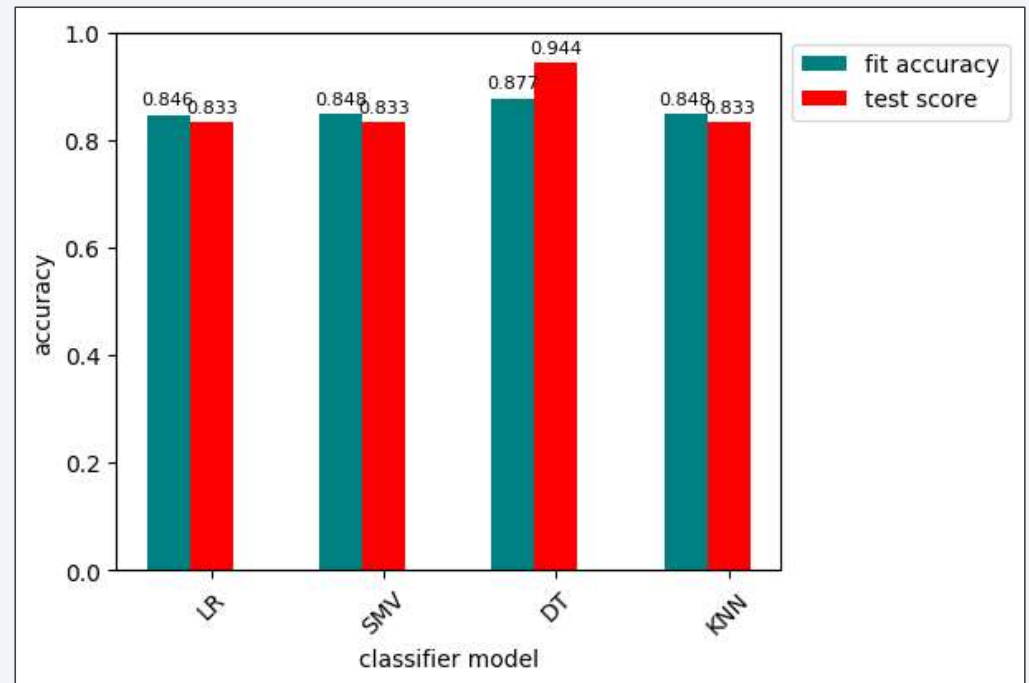
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Using ML classifier models to predict SpaceX launched Falcon 9 landing success, 4 models were tried: Logistic Regression (LR), Support Machine Vector (SMV), Decision Tree (DT) and K-Nearest Neighbor Classifier (KNN)

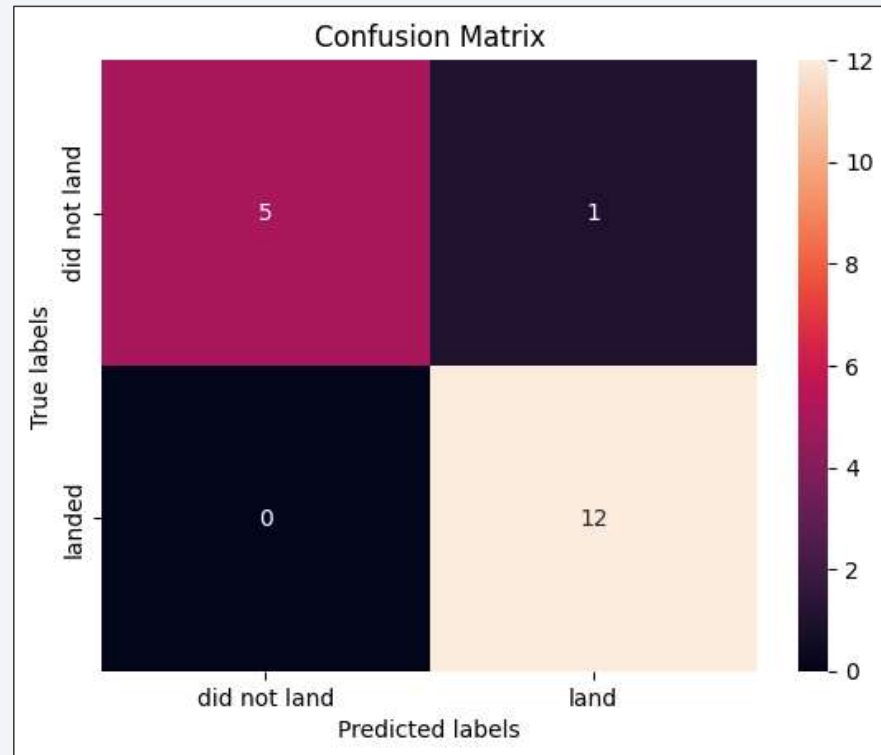
The results of the fit accuracy and model score from the test data are shown, demonstrating that decision tree worked the best (~0.88 accuracy, 0.94 score) for fitting/predicting outcomes, while the other models showed mostly the same response at ~0.85



Confusion Matrix

Confusion matrix for the Decision Tree classifier model used in predicting SpaceX Falcon 9 launch landing success.

It is seen the model predicts true positives/negatives, with only a single missed false positive.



Conclusions

- SpaceX has, over a decade, improved its landing capabilities – it will be increasingly difficult to expect failed landings
- There is a trend towards larger payloads ($> 10,000$ kg) which in turn changes the geographic launch pad which can handle the payload, and the booster that can be used
- These higher payload launches have higher success of landing (~80%)
- The launch sites for these larger payloads are all in Florida, closer to the equator, and generally appear to be for VLEO orbits or for deliveries to the ISS
- ML can give accuracy in the ~90%+ range for predicting landing success/failure, but we must carefully assess for false positives

Appendix

The following data sets and python code was generated for undertaking this project:

- map_distance2.html
- map_launch_class.html
- SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb
- dashboard_script.py
- my_data1.db
- dataset_part_1.csv
- dataset_part_2.csv
- dataset_part_3.csv
- space_web_scrape.csv
- spacex_launch_dash.csv
- jupyter-labs-eda-dataviz-v2.ipynb
- jupyter-labs-eda-sql-coursera_sqlite.ipynb
- jupyter-labs-spacex-data-collection-api-v2.ipynb
- jupyter-labs-webscraping.ipynb
- lab-jupyter-launch-site-location-v2.ipynb
- labs-jupyter-spacex-Data wrangling-v2.ipynb'
- map.html
- map_distance.html
- map_launch_sites.html
- spacex_launch_geo.csv

Thank you!

